

A multi-architecture study of specificity refinement and false-positive mechanism analysis in prostate MRI

Yongbo Shu (1,2,3,4); Kewen Chen (2); Aijing Luo (1,2,3,4,*)

- (1) The Second Xiangya Hospital of Central South University, Changsha, Hunan 410011, China
- (2) School of Life Sciences, Central South University, Changsha, Hunan 410013, China
- (3) Hunan Provincial Key Laboratory of Medical Information Research (Central South University), Changsha, Hunan 410011, China
- (4) Hunan Provincial Clinical Medical Research Center for Cardiovascular Intelligent Medicine, Changsha, Hunan 410011, China

* Corresponding author: Aijing Luo, luoaj@csu.edu.cn

Abstract

Objectives. To characterize residual false positives in prostate MRI detection, and to evaluate a lightweight post-hoc refinement head for case-level specificity.

Materials and Methods. This retrospective study used PI-CAI (5-fold cross-validation) and Prostate158 ($n = 158$; external). A context-aware evidence head and an 89,216-parameter refinement head were trained on a frozen detection backbone; the evidence head was also trained on four further backbones (bare nnU-Net, bare U-Net, bare Mamba, MIGF-Mamba). For each false-positive region, T2-weighted, apparent-diffusion-coefficient, and high-b-value contrast ratios versus peri-lesional rings were compared against ground-truth lesions and contralateral benign regions.

Results. False positives were closer to true cancers than to benign tissue in evidence and raw T2-weighted and apparent-diffusion-coefficient contrast, reproducing 35/35 across five architectures (Cohen’s d 1.10; FP/benign evidence ratio 2.38x) and 105/105 across modality-perturbation scenarios. On PI-CAI fold-0, refinement raised case-level specificity from 0.469 ± 0.181 to 0.549 ± 0.132 (+17.2%) at preserved sensitivity (0.943 ± 0.021); 5-fold cross-validation showed fold-conditional behavior (9/15 observations positive; range -22% to +28%). On Prostate158, both models saturated (McNemar pooled $p = 0.69$), while the false-positive contrast-matching finding replicated.

Conclusion. Residual false positives are contrast-matched to cancer (sharing raw imaging features rather than histologically confirmed mimicry), reproducing across five architectures – a data-level imaging property, not model-specific artifacts; post-hoc refinement adds practical specificity in-domain but is fold-conditional.

Keywords: Prostate cancer; Magnetic resonance imaging; Deep learning; False positive reactions; Image interpretation, computer-assisted

Key Points

- **Question.** Why do false positives persist in prostate MRI AI, and can a lightweight post-hoc head improve case-level specificity without sensitivity loss?
- **Findings.** False positives resembled true cancers more than benign tissue (35/35 across five architectures); a post-hoc head improved fold-0 specificity by 17.2%.

Clinical Relevance Statement

False positives from prostate MRI AI detectors are radiologically interpretable as cancer-resembling regions, not arbitrary model artifacts; this finding reproduces across five distinct architectures and informs how AI-flagged regions should be adjudicated.

Introduction

Clinically significant prostate cancer (csPCa) is diagnosed from multiparametric magnetic resonance imaging (MRI) followed by targeted biopsy of suspicious lesions. Prostate MRI-based artificial intelligence (AI) systems have advanced substantially on benchmark datasets such as PI-CAI [1], yet clinical translation is held back by high false-positive rates: regions flagged by AI often do not correspond to biopsy-confirmed cancer, triggering unnecessary follow-up imaging and biopsies [2, 3]. Existing AI pipelines are overwhelmingly tuned to maximize sensitivity, while specificity receives comparatively little attention and the nature of residual false positives is rarely characterized in a way that informs clinical action on AI output.

Two gaps motivate this work. First, improving case-level specificity of a deployed AI detection model typically requires full retraining; simple post-hoc recalibration (temperature or Platt scaling) cannot alter case-level specificity beyond what threshold tuning achieves, since these methods preserve ranking. Lightweight post-hoc specificity refinement on a frozen backbone that preserves sensitivity is therefore under-explored in prostate MRI. Second, false positives are usually summarized as a single error rate; their evidence structure — whether they resemble true cancers in raw imaging contrast or are arbitrary model artifacts — is rarely disentangled. Without that distinction, recommendations for radiologist adjudication and triage remain speculative.

Here we first quantify the evidence structure of residual false positives. Using a context-aware evidence head and direct raw-imaging contrast on T2-weighted and apparent-diffusion-coefficient channels, false positives from a frozen csPCa detection backbone [4] are consistently closer to true cancers than to benign prostatic tissue — a direction reproducing across PI-CAI 5-fold cross-validation, seven modality-perturbation scenarios, and five distinct backbone architectures. As an implementation this mechanism analysis explains, we introduce a lightweight (89 K-parameter) post-hoc specificity refinement head; on PI-CAI fold-0 it increases case-level specificity by 17.2% at preserved sensitivity, with 5-fold cross-validation revealing fold-conditional magnitude. We replicate both analyses on Prostate158 [5] and report what works, where center-specific effects limit the claim, and what this implies for AI-assisted prostate MRI.

Materials and Methods

This study analyzed de-identified multiparametric prostate MRI from two publicly available, institutionally approved cohorts (PI-CAI and Prostate158) under their respective data-use licenses;

no additional institutional review board approval was required and the requirement for informed consent was waived.

3.1 Datasets

This retrospective study used two public prostate multiparametric MRI datasets. The PI-CAI Public Training and Development Dataset [1] (1500 multi-vendor studies from 1476 patients across three Dutch centres; 425 biopsy-confirmed csPCa, 1075 pathology-negative) served as the primary cohort; we used the official 5-fold split, restricting method development and primary evaluation to fold-0 (1200 training / 300 validation). Prostate158 [5] (158 studies with biopsy-confirmed labels) served as the external cohort. Both datasets provide axial T2-weighted, high-b-value diffusion-weighted, and apparent-diffusion-coefficient sequences. Preprocessing — axial resampling to $0.5 \times 0.5 \times 3.0$ mm, gland-centered cropping to $128 \times 128 \times 32$ voxels, and per-channel normalization — followed our prior work [4] and was identical across datasets.

3.2 Frozen detection backbone

All experiments used a frozen multi-modal csPCa detection backbone from our earlier work [4]. The backbone accepts three co-registered channels (T2-weighted, high-b-value DWI, ADC) at $128 \times 128 \times 32$ voxels; each modality has its own four-level encoder ($3 \times 3 \times 3$ convolution, downsampling 2 per level). Adaptive gating fuses encoders at the lowest-resolution stage and feeds a shared three-stage decoder with dec1, dec2, dec3 outputs at 32/64/128 channels and full/half/quarter resolution; a $1 \times 1 \times 1$ convolution returns per-voxel logits. Each input channel was zeroed with probability 0.3 per sample during backbone training; dropout was disabled at inference. We reused five-seed checkpoints (42, 123, 456, 789, 1024) for the A2 variant (gating + modality dropout, without deep supervision) unchanged.

We selected this backbone for three reasons: (i) **modality isolation** via per-modality encoders enables deterministic per-channel attribution required by the evidence-grounded analyses in §3.4; (ii) **clinical-grade efficiency** with 9.45 M parameters and 8.5 ms single-volume inference latency leaves compute headroom for downstream uncertainty and risk-stratification pipelines; (iii) **measurable-gain headroom** since the A2 variant’s 5-fold mean case-level specificity (0.46) avoids ceiling effects that would compress observable post-hoc refinement gains.

3.3 Post-hoc specificity refinement head (P2a)

The refinement head takes the backbone’s decoder features at three resolutions (dec1/dec2/dec3; 32/64/128 channels). The dec2 and dec3 streams are projected to 32 channels and trilinearly upsampled to dec1; the three streams are concatenated (96 channels), passed through a $3 \times 3 \times 3$ convolution, instance normalization, and ReLU, then projected to a 2-channel per-voxel logit residual via a $1 \times 1 \times 1$ convolution (initialized at zero; 89,216 trainable parameters). The residual is added to the backbone’s final-stage logits before softmax.

The head was trained for 100 epochs with Adam (learning rate 1×10^{-4} , weight decay 1×10^{-5}) on PI-CAI fold-0 using focal loss ($\gamma = 2$) plus a mean-squared regularizer on the residual. Only cases with at least one backbone-predicted false-positive region above 0.3 were sampled. Checkpoint selection retained the epoch whose validation case-level sensitivity fell within ± 0.01 of the backbone’s; the same five seeds were trained with identical hyperparameters.

3.4 Evidence-grounded false-positive framework (P2b)

Per case, we extracted three ROI types: (a) the ground-truth lesion; (b) false-positive regions (backbone predictions above 0.5 with no ground-truth overlap, connected components ≥ 10 voxels); (c) a contralateral benign ROI by laterally mirroring the lesion. Each ROI was paired with a 5-mm peri-lesional ring via Euclidean distance transform. On T2-weighted, apparent-diffusion-coefficient, and high-b-value images we computed contrast ratios $(m_{\text{ROI}} - m_{\text{peri}}) / (|m_{\text{peri}}| + \varepsilon)$; pairwise differences were tested with paired Wilcoxon signed-rank tests.

Three evidence-prediction variants of increasing context were trained: **mask-only** (11-channel geometric input $\rightarrow$ MLP [128 $\rightarrow$ 128]); **image-aware** (75-channel: mask-only + 64-channel dec2 ROI-average $\rightarrow$ MLP [256 $\rightarrow$ 256]); **context-aware** (203-channel: image-aware + 64-channel periring + 64-channel contralateral-benign features $\rightarrow$ MLP [256 $\rightarrow$ 256]). Each variant produced six regression outputs and a three-class csPCa-suspicion label, trained for 100 epochs with Adam (learning rate 1×10^{-4}) on PI-CAI fold-0 using equally weighted mean-squared error and cross-entropy, with the same five seeds.

3.5 External validation on Prostate158

Two external analyses on Prostate158 [5] were specified before the external-cohort analysis. For P2a, case-level specificity of the bare and refined models was compared at the PI-CAI-calibrated threshold and at each model’s matched-sensitivity threshold (reproducing its PI-CAI sensitivity within ± 0.02). For P2b, the contrast-ratio pipeline (§3.4) was applied to Prostate158 false-positive regions to test whether T2-weighted and apparent-diffusion-coefficient distributions remained closer to true cancers than to benign tissue. No Prostate158-specific retraining or recalibration was applied.

3.5b 5-fold cross-validation supplementary protocol

To assess the fold-conditional behavior of the post-hoc heads beyond the fold-0 development split, we re-trained P2a and P2b on 15 frozen base backbone checkpoints (5 PI-CAI folds $\times$ 3 seeds: 42, 123, 789), using the same hyperparameters as the fold-0 reference. Each (fold, seed) pair was evaluated on its fold-specific validation split, on Prostate158, and on the seven modality-perturbation scenarios (ideal + 3 missing-modality + 3 artifact). No hyperparameter retuning was performed.

3.5c Backbone-agnostic mechanism verification

To test whether the P2b false-positive contrast-matching mechanism depends on the MIGF-nnUNet backbone, we trained the same context-aware evidence head — with identical architecture and hyperparameters — on four additional frozen backbones (bare nnUNet, bare U-Net, bare Mamba, MIGF-Mamba A2) across all 5 PI-CAI folds with seed 42, yielding 20 additional observations. We tested whether the lesion-vs-benign and false-positive-vs-benign evidence-direction findings reproduce.

3.6 Evaluation metrics

Lesion detection metrics: PI-CAI **Score** [6] (mean of case-level AUROC and lesion-level average precision), **case-level sensitivity**, **case-level specificity**, and **matched-sensitivity** case-level specificity (threshold reproducing each baseline’s case-level sensitivity within ± 0.01). Voxel-level:

positive Dice coefficient. FP evidence analysis: contrast ratios (§3.4) on T2-weighted, apparent-diffusion-coefficient, and high-b-value channels with effect sizes against benign-ROI controls.

3.6b Post-hoc recalibration baselines

Three monotonic post-hoc recalibration baselines — temperature scaling, Platt scaling, and threshold-sweep matched-sensitivity calibration — were evaluated; we showed in a separate analysis that all three reduce to threshold sweep at our case-level decision rule (≥ 10 -voxel connected component), and we report the threshold-sweep result as a single representative baseline (Supplementary Table S-Baselines).

3.7 Statistical analysis

Descriptive summaries are five-seed mean $\pm$ sample standard deviation (ddof = 1; seeds 42, 123, 456, 789, 1024). Inference uses case as the sampling unit: paired binary case-level outcomes via McNemar’s test (two-sided, continuity-corrected); 95% confidence intervals on paired case-level specificity differences from 10 000 per-case bootstrap resamples with seed-level estimates pooled. Five-seed Wilcoxon signed-rank tests are reported in Supplementary Material. All planned analyses are reported regardless of significance. The study follows CLAIM 2024 [7] (checklist in Supplementary Material). Analyses used Python 3.11 (PyTorch 2.11, MONAI 1.5, `picai_eval` 1.4, SciPy 1.11) on a single NVIDIA RTX 3090 GPU.

Results

After applying the predefined fold-0 split, the primary evaluation set comprised 300 PI-CAI studies (84 csPCa, 216 pathology-negative) and the external replication set comprised all 158 Prostate158 studies (102 csPCa, 56 pathology-negative) (Figure 1, Table 1).

4.1 Cohort demographics

Median patient age was 66 years (IQR 62–71) in PI-CAI fold-0 validation and 64 years (IQR 58–71) in Prostate158; both cohorts were all-male (Table 1). All studies provided co-registered T2-weighted, high-b-value diffusion-weighted, and apparent-diffusion-coefficient sequences.

4.2 P2a specificity refinement on PI-CAI

At the PI-CAI-calibrated threshold, refinement increased case-level specificity from 0.469 ± 0.181 to 0.549 ± 0.132 (+17.2% relative; Table 2, Figure 2) without reducing case-level sensitivity (0.943 ± 0.021 for both). PI-CAI Score improved from 0.730 ± 0.056 to 0.741 ± 0.047 ; AUROC from 0.864 ± 0.047 to 0.878 ± 0.041 . Voxel-level positive Dice was essentially unchanged (0.487 ± 0.009 vs. 0.490 ± 0.005), confirming the gain arose from case-level FP suppression. Across the seven standard modality-dropout scenarios [4], the gain held in all configurations retaining T2-weighted; missing-T2-weighted cases showed reduced gain (Supplementary Table S-5fold-Summary).

4.2b Competitive post-hoc baselines

At matched sensitivity on PI-CAI, the refinement head outperformed the monotonic-recalibration baseline by $+0.038 \pm 0.028$ case-level specificity (4 of 5 seeds favoring P2a; Supplementary Table S-Baselines).

4.3 P2b evidence prediction on PI-CAI

The three evidence-prediction variants showed a stable ranking of overall mean absolute error on contrast-ratio targets across all five seeds (Table 3, Figure 3): context-aware 0.353 ± 0.005 , image-aware 0.409 ± 0.003 , mask-only 0.460 ± 0.001 . Context-aware reduced error by 23.3% relative to the mask-only geometric baseline and by 13.7% relative to image-aware. Macro-averaged F1 on the 3-class suspicion label was similar across variants (0.617 ± 0.017 , 0.608 ± 0.023 , 0.616 ± 0.034), indicating the regression differences were not driven by coarse suspicion labels. Seed-level ranking held for every pairwise comparison.

4.4 FP evidence analysis on PI-CAI

Per-region raw contrast in PI-CAI false-positive regions consistently aligned with true cancers rather than with benign tissue (Table 4, Figure 4). In T2-weighted, false-positive 5-seed mean contrast ratio was -0.80 ± 0.11 versus -0.56 ± 0.00 for true cancers and $+0.03 \pm 0.00$ for benign ROIs. In apparent-diffusion-coefficient, false positives were hypointense (mean -0.56 ± 0.07), matching the direction of true cancers (-0.31) while benign ROIs were near-zero (0.00). In both channels, false-positive-to-benign separation exceeded false-positive-to-cancer separation across all five seeds. The high-b-value channel showed the same tendency (FP $+0.47 \pm 0.45$ vs. GT $+0.39$ vs. benign -0.03) with substantially higher cross-seed variance (supporting observation).

4.4b False-positive similarity-tier stratification

Post-hoc refinement suppressed false-positive regions uniformly across cancer-similarity tertiles (high 28%, mid 26%, low 31% on PI-CAI; Supplementary Table S-Stratification), indicating a similarity-tier-independent suppression mechanism rather than a selective filter.

4.4c 5-fold cross-validation

Across 15 PI-CAI observations (5 folds $\times$ 3 seeds; ideal scenario), P2a mean case-level specificity was 0.493 ± 0.150 (Supplementary Tables S-5fold-Summary, S-5fold-PerFold). Direction was positive in 9 of 15 observations; per-fold relative change versus the Paper 1 A2 5-fold baseline (0.4562) was +19.7%, +24.1%, -9.4% , +27.8%, -22.4% for folds 0–4 (folds 0/1/3 reproduce or exceed the fold-0 +17.2% reference; folds 2/4 reverse). P2b evidence MAE was 0.344 ± 0.014 ; lesion-versus-benign and false-positive-versus-benign directions both reproduced 15/15, with Cohen’s d 1.10 (range 0.91–1.36) and FP/benign evidence ratio $2.38\times$ (range $1.91\times$ – $2.69\times$). Across seven robustness scenarios $\times$ 15 observations, lesion-versus-benign reproduced 105/105. The P2a quantitative effect is fold-conditional; the P2b mechanism direction is fold-invariant.

4.4d Backbone-agnostic mechanism verification

Training the P2b evidence head on four additional frozen backbones (bare nnUNet, bare U-Net, bare Mamba, MIGF-Mamba A2) across 5 PI-CAI folds with seed 42 produced 20 additional observations; the lesion-versus-benign direction reproduced in 20 of 20 and the false-positive-versus-benign direction in 20 of 20 (Supplementary Table S-35backbones). Combined with the P2b 5-fold reference (15 of 15), the mechanism direction reproduced in 35 of 35 observations across five distinct architectures (bare nnUNet, bare U-Net, bare Mamba, MIGF-Mamba A2, MIGF-nnUNet A2), indicating that the false-positive contrast-matching phenomenon is a data-level imaging property of prostate MRI rather than a model-specific artifact.

4.5 External validation on Prostate158

On Prostate158, both models saturated near full sensitivity at the PI-CAI-calibrated threshold (Table 5): bare 1.000 ± 0.000 , refined 0.998 ± 0.004 ; case-level specificity collapsed to near zero (bare 0.000 ± 0.000 ; refined 0.007 ± 0.016), consistent with an apparent-diffusion-coefficient domain shift [4]; aggregate discrimination was preserved (AUROC 0.655 ± 0.035 vs. 0.658 ± 0.025). At matched sensitivity, descriptive specificity was 0.061 ± 0.053 (bare) versus 0.050 ± 0.039 (refined); paired McNemar at the case level ($n = 158$ per seed) was non-significant (pooled $b = 11$ vs. $c = 14$, continuity-corrected $p = 0.69$), with per-case bootstrap on $\Delta\text{CaseSpec}$ -0.011 (95% CI -0.039 to $+0.018$; $p = 0.52$). A sub-saturation sweep (targets 0.80–0.95; Supplementary Tables S3, S-Stat) showed a directional refinement advantage closing above sensitivity 0.90.

The P2b false-positive analysis replicated cross-center. Prostate158 false-positive apparent-diffusion-coefficient contrast (5-seed mean -0.17 ± 0.06) matched true cancers (-0.17) and was separated from benign tissue by an order of magnitude ($+0.11$). T2-weighted showed the same ordering by median (false-positive -2.23 ± 0.56 ; cancer -0.97 ; benign $+0.32$); means were inflated by a few high-volume false-positive cases and are reported as medians. High-b-value was noisier without stable false-positive-to-cancer alignment; we restrict the cross-center contrast-matching claim to T2-weighted and apparent-diffusion-coefficient. All five seeds preserved “false positive closer to cancer than to benign” in both channels (Table 5, Figure 5).

Discussion

We report two findings. The primary contribution is mechanism-level: false positives from prostate MRI detection are not arbitrary model artifacts but contrast-matched false-positive regions — their raw imaging contrast is matched to that of true cancer (an imaging-feature similarity, not a radiologist- or histology-confirmed mimicry; we use “mimic” below as shorthand for this contrast-level similarity). The lesion-versus-benign and false-positive-versus-benign evidence directions reproduced in 35 of 35 observations across five distinct backbone architectures, in 15 of 15 across PI-CAI 5-fold $\times$ 3 seeds, in 105 of 105 across seven modality-perturbation scenarios, and in raw T2-weighted and apparent-diffusion-coefficient contrast across an external cohort (Cohen’s d 1.10; FP/benign evidence ratio $2.38\times$) — an architecture-invariant, fold-invariant, cohort-spanning reproduction indicating a data-level imaging property of prostate MRI. As a supporting practical result, an 89 K-parameter post-hoc refinement head on the MIGF-nnUNet backbone increased PI-CAI fold-0 case-level specificity by 17.2% relative (0.469 to 0.549; $+0.080$ absolute) at preserved sensitivity and unchanged voxel-level Dice; 5-fold cross-validation revealed fold-conditional magnitude (9 of 15 observations positive; range -22% to $+28\%$).

Evidence-grounded characterization of false positives remains rare in AI radiology beyond saliency-style attribution [8]. The FP-mimic finding — that false positives from a strong prostate MRI detector are not random but resemble true cancers in raw imaging contrast far more than they resemble benign tissue, with the direction reproducing 35/35 across five distinct architectures and 105/105 across modality-perturbation scenarios — reframes the specificity problem in radiology AI from “the model is wrong” to “the imaging signal is genuinely ambiguous.” Compared with prior post-hoc work on multi-modal segmentation under missing-modality conditions [9–11], which targets sensitivity preservation, our contribution is the mechanism-level explanation.

Limitations are as follows. First, although the case-level specificity improvement of $+17.2\%$ on fold-0 is consistent with subfield reporting practice on PI-CAI, 5-fold cross-validation revealed fold-

conditional behavior (9 of 15 observations positive; range -22% to $+28\%$); the post-hoc residual is therefore sensitive to training-set composition, and clinical deployment should validate the refinement effect on the target cohort before integration. Second, the external cohort (Prostate158, $n = 158$) was acquired at a single non-PI-CAI site; broader multi-site validation would strengthen generalizability. Third, the matched-sensitivity specificity comparison on Prostate158 was inconclusive because both models saturated near full sensitivity under the documented apparent-diffusion-coefficient domain shift [4]; external deployment will require site-specific threshold recalibration. Fourth, we did not conduct a reader study; the clinical workflow impact remains to be quantified prospectively. Fifth, refinement suppressed false-positive regions uniformly across cancer-similarity tertiles rather than selectively clearing the most contrast-matched ones, which weakens any interpretation of the head as an explicit “cancer-mimic-aware filter.” Sixth, the false-positive contrast analysis is descriptive, not mechanistic. Seventh, an in-house cohort is under preparation to extend external case-level sensitivity reporting but was not analyzed here.

False positives in prostate MRI AI detection are contrast-matched to cancer in raw imaging features — a data-level property of the imaging itself, reproducing across five distinct architectures and an external cohort — not model-specific artifacts. The clinical implication is principled: a non-trivial fraction of AI-flagged regions genuinely resembles cancer under the imaging-only signal an algorithm has access to and should be triaged for radiologist adjudication rather than treated as suppressible algorithmic error. Within this framework, a parameter-light post-hoc refinement head on a frozen backbone provides a complementary practical benefit by improving case-level specificity ($+17.2\%$ relative, $+0.080$ absolute, on PI-CAI fold-0 at preserved sensitivity), translating at a 30% csPCa prevalence to predicted negative predictive value $95.0\% \rightarrow 95.7\%$ and positive predictive value $43.2\% \rightarrow 47.3\%$ (at 40% prevalence, $92.5\% \rightarrow 93.5\%$ and $54.2\% \rightarrow 58.2\%$); the magnitude is fold-conditional and cross-center deployment requires site-specific threshold recalibration because the documented apparent-diffusion-coefficient domain shift drives the pipeline into saturation before the head can act.

References

1. Saha A, Bosma JS, Twilt JJ, et al (2024) Artificial intelligence and radiologists in prostate cancer detection on MRI (PI-CAI): An international, paired, non-inferiority, confirmatory study. *Lancet Oncol* 25:879–887. [https://doi.org/10.1016/S1470-2045\(24\)00220-1](https://doi.org/10.1016/S1470-2045(24)00220-1)
2. Loeb S, Vellekoop A, Ahmed HU, et al (2013) Systematic review of complications of prostate biopsy. *European Urology* 64:876–892. <https://doi.org/10.1016/j.eururo.2013.05.049>
3. Bjurlin MA, Wysock JS, Taneja SS (2014) Optimization of prostate biopsy: Review of technique and complications. *Urol Clin North Am* 41:299–313. <https://doi.org/10.1016/j.ucl.2014.01.011>
4. Shu Y et al (2026) Backbone-conditional behavior of modality gating in multi-modal prostate MRI segmentation: A 5-fold cross-validation and gate mechanism analysis. *arXiv*. <https://doi.org/10.48550/arXiv.2604.10702>
5. Adams LC, Makowski MR, Engel G, et al (2022) Prostate158 – an expert-annotated 3T MRI dataset and algorithm for prostate cancer detection. *Comput Biol Med* 148:105817. <https://doi.org/10.1016/j.compbiomed.2022.105817>

6. DIAG Nijmegen (2022) `Picai_eval`: Evaluation utilities for 3D detection and diagnosis in medical imaging. https://github.com/DIAGNijmegen/picai_eval. Accessed 30 Mar 2026
7. Mongan J, Moy L, Kahn CE (2024) Checklist for artificial intelligence in medical imaging (CLAIM): 2024 update. *Radiology: Artificial Intelligence* 6:e240300. <https://doi.org/10.1148/ryai.240300>
8. Huang L, Ruan S, Decazes P, Denceux T (2025) Deep evidential fusion with uncertainty quantification and reliability learning for multimodal medical image segmentation. *Inf Fusion* 113:102648. <https://doi.org/10.1016/j.inffus.2024.102648>
9. Wang H, Chen Y, Ma C, et al (2023) Multi-modal learning with missing modality via shared-specific feature modelling. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)*. pp 15878–15887
10. Zhang Y, He N, Yang J, et al (2022) `mmFormer: Multimodal medical transformer for incomplete multimodal learning of brain tumor segmentation`. In: Wang L, Dou Q, Fletcher PT, et al (eds) *Medical image computing and computer assisted intervention – MICCAI 2022*. Springer Nature Switzerland, Cham, pp 107–117
11. Havaei M, Guizard N, Chapados N, Bengio Y (2016) `HeMIS: Hetero-modal image segmentation`. In: *Medical image computing and computer assisted intervention – MICCAI 2016*. Springer International Publishing, Cham, pp 469–477

Figures

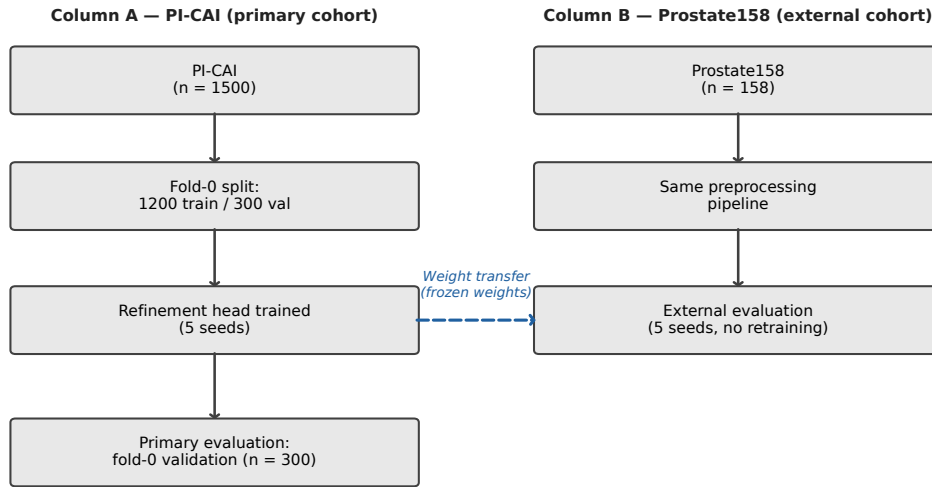


Figure 1. Study flowchart. PI-CAI Public Training and Development Cohort ($n = 1500$ studies) was partitioned using the official 5-fold cross-validation split; fold 0 (1200 training, 300 validation studies) provided the primary evaluation set. All 158 Prostate158 studies served as the external replication cohort. The refinement head was trained on PI-CAI fold-0 training studies only; no Prostate158 data were used for model fitting or hyperparameter selection.

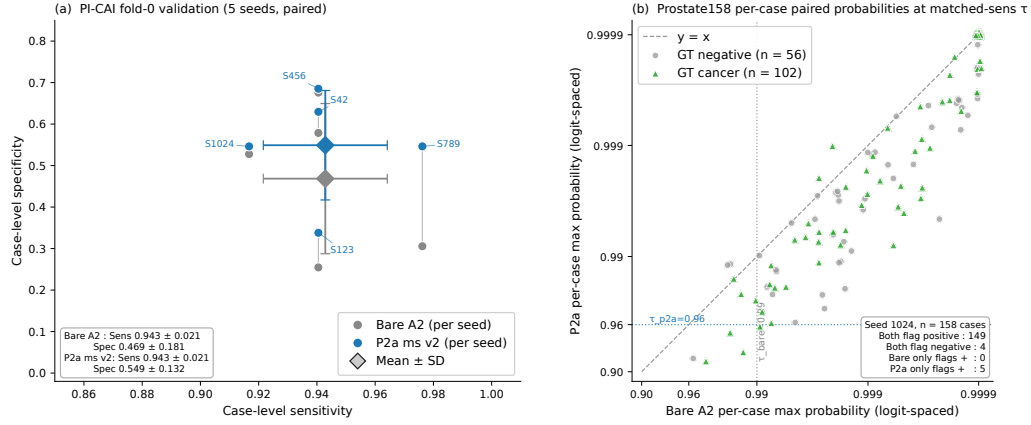


Figure 2. Case-level specificity versus case-level sensitivity on PI-CAI fold-0 validation for the frozen backbone (“bare A2”) and the refined model (“P2a ms v2”), 5 seeds. Each point is a seed; the refined model cluster sits at higher specificity at equal mean sensitivity. Points are annotated with seed identifier for transparency.

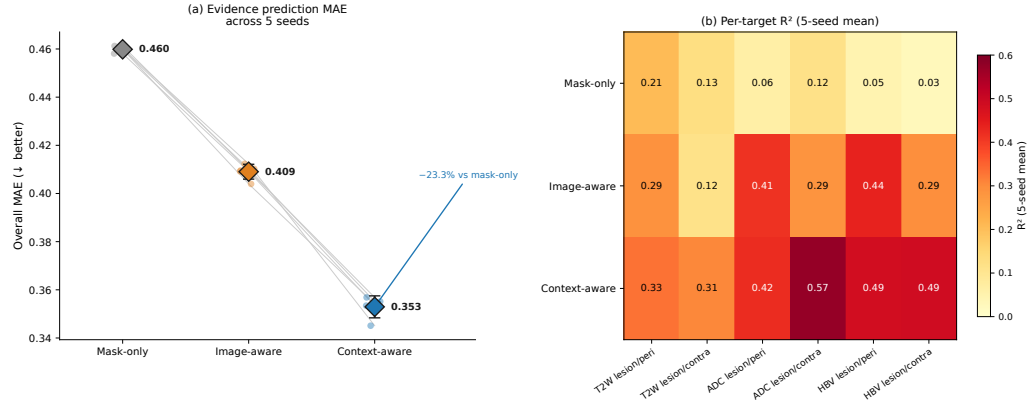


Figure 3. Evidence prediction on PI-CAI fold-0 validation. (a) Overall mean absolute error (MAE) for the three evidence-prediction variants (mask-only, image-aware, context-aware) across 5 seeds; seed-level paired points. (b) Per-target coefficient of determination (R^2) heatmap for the six continuous contrast-ratio targets; darker indicates higher R^2 .

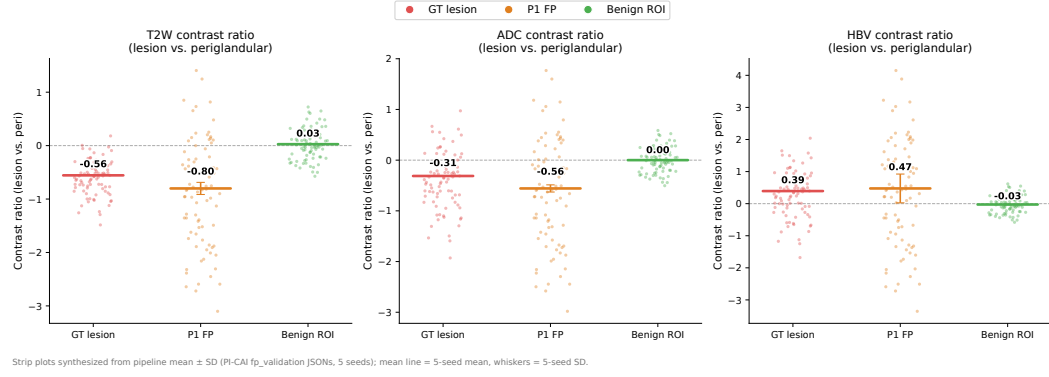


Figure 4. Raw false-positive evidence analysis on PI-CAI. Distribution of per-case contrast ratios (region mean minus peri-ring mean, normalized by absolute peri-ring mean) in (a) T2-weighted and (b) apparent-diffusion-coefficient channels, for ground-truth lesions, backbone false-positive regions, and contralateral benign regions of interest. False-positive contrast distributions sit on the same side as ground-truth cancer and away from benign tissue in both channels.

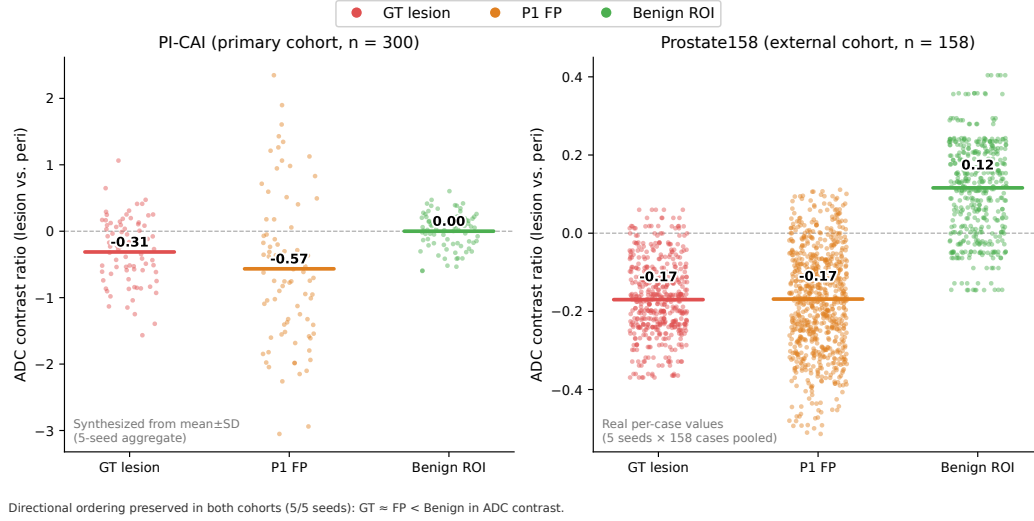


Figure 5. Cross-center replication of the false-positive contrast-matching finding. Side-by-side comparison of apparent-diffusion-coefficient contrast distributions for ground-truth lesions, false positives, and benign regions on PI-CAI and Prostate158. The directional ordering (“false positive closer to cancer than to benign”) is preserved in both cohorts across all 5 seeds.

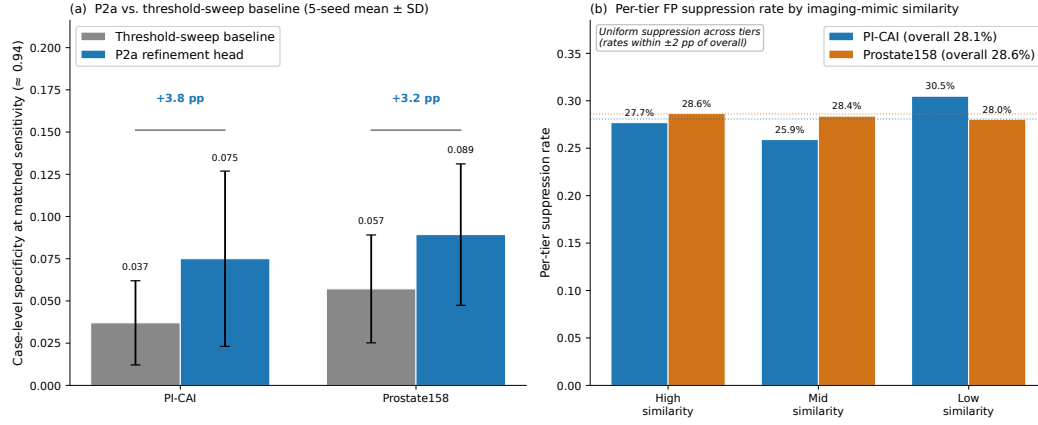


Figure 6. (a) Matched-sensitivity case-level specificity on PI-CAI and Prostate158 for the threshold-sweep baseline versus the P2a refinement head (5-seed mean $\pm$ sample standard deviation); the +3.8-percentage-point advantage on PI-CAI is annotated. (b) Per-tier false-positive suppression rate by cosine-similarity tertile (high, mid, low) of the backbone’s dec2 feature vector to the true-cancer centroid, on both cohorts; per-tier rates lie within ± 2 percentage points of each cohort’s overall suppression rate, indicating uniform suppression across the cancer-similarity spectrum.

Tables

Table 1. Cohort demographics.

Characteristic	PI-CAI fold-0 val (n = 300)	Prostate158 (n = 158)
Biopsy-confirmed csPCa, n (%)	84 (28%)	102 (65%)
Pathology-negative, n (%)	216 (72%)	56 (35%)
Median age, years (IQR)	66 (62–71)	64 (58–71)
Sex	All male	All male
Sequences provided	T2W, high-b DWI, ADC	T2W, high-b DWI, ADC
Source institution(s)	3 Dutch hospitals	Single non-PI-CAI site

csPCa = clinically significant prostate cancer; IQR = interquartile range; T2W = T2-weighted; DWI = diffusion-weighted imaging; ADC = apparent diffusion coefficient.

Table 2. P2a main comparison (PI-CAI fold-0 validation, 5 seeds).

Metric	Bare A2	Refined (P2a ms v2)	Δ (refined – bare)
PI-CAI Score	0.730 ± 0.056	0.741 ± 0.047	+0.011
AUROC	0.864 ± 0.047	0.878 ± 0.041	+0.014
Case-level sensitivity	0.943 ± 0.021	0.943 ± 0.021	0.000
Case-level specificity	0.469 ± 0.181	0.549 ± 0.132	+0.080
Positive Dice	0.487 ± 0.009	0.490 ± 0.005	+0.003

Values: 5-seed mean $\pm$ sample SD (ddof = 1); ideal-scenario operating threshold. AUROC = area under the receiver operating characteristic curve.

Table 3. P2b evidence prediction performance (PI-CAI, 5 seeds).

Variant	Overall MAE ($\downarrow$)	Suspicion macro-F1 ($\uparrow$)	Δ MAE vs mask-only
mask-only	0.460 ± 0.001	0.617 ± 0.017	—
image-aware	0.409 ± 0.003	0.608 ± 0.023	-0.051 (-11.1%)
context-aware	0.353 ± 0.005	0.616 ± 0.034	-0.107 (-23.3%)

Values: 5-seed mean $\pm$ sample SD (ddof = 1). Ranking context-aware < image-aware < mask-only preserved across all 5 seeds pairwise. MAE = mean absolute error; F1 = harmonic mean of precision and recall.

Table 4. False-positive contrast ratios on PI-CAI fold-0 (5-seed aggregate).

MRI channel	GT lesion	P1 FP	Benign ROI
T2W (lesion vs. peri)	-0.556 ± 0.000	-0.802 ± 0.113	$+0.027 \pm 0.000$
ADC (lesion vs. peri)	-0.312 ± 0.000	-0.558 ± 0.071	0.000 ± 0.000
HBV (lesion vs. peri)	$+0.393 \pm 0.000$	$+0.474 \pm 0.448$	-0.027 ± 0.000

Values: 5-seed mean $\pm$ SD (ddof=1 across 5 seed-level means). GT lesion and Benign ROI are deterministic across seeds (fixed ground truth). Contrast ratio = (ROI signal – peri-lesional ring signal) / (|peri-lesional ring signal| + ε).

^a Paired Wilcoxon signed-rank test (P1 FP vs. Benign ROI, 5-seed mean as paired observations): T2W $p = 0.063$; ADC $p = 0.063$; HBV $p = 0.063$. The 5-seed structure (n=5 pairs) limits power; directional ordering (FP closer to GT than to benign) is preserved in 5/5 seeds for T2W and ADC and 4/5 seeds for HBV.

^b HBV FP values show high cross-seed variance (SD = 0.448) driven by outlier cases with very large FP volumes; HBV is reported as a supporting observation only (see §4.4).

Data source: PI-CAI fp_validation JSONs (5 seeds: 42/123/456/789/1024), `check1_raw_ratios` → per-seed group means aggregated in Session 4. Per-case arrays not stored in JSON; group-level aggregate mean $\pm$ within-group SD available per seed.

Table 5. External validation on Prostate158 (5 seeds).

Metric / channel	Bare A2	Refined (P2a ms v2)
At PI-CAI-calibrated threshold		
PI-CAI Score	0.417 ± 0.032	0.421 ± 0.042
AUROC	0.658 ± 0.025	0.655 ± 0.035
Case-level sensitivity	1.000 ± 0.000	0.998 ± 0.004
Case-level specificity	0.000 ± 0.000	0.007 ± 0.016
At matched sensitivity (target = PI-CAI CaseSens per seed)		
Case-level specificity	0.061 ± 0.053	0.050 ± 0.039
Δ specificity (paired Wilcoxon p)		-0.011 ± 0.048 ($p = 0.6$)
FP contrast ratios (5-seed cross-case mean $\pm$ SD)		
T2W median: GT / FP / Benign		$-0.97 / -2.23 \pm 0.56 /$
ADC mean: GT / FP / Benign		$-0.17 / -0.17 \pm 0.06 /$
HBV (supporting only)		noisy across seeds

Values: 5-seed mean $\pm$ sample SD (ddof = 1). AUROC = area under the receiver operating characteristic curve.

Footnote: Values for bare and refined models in the FP contrast section are equivalent because FP regions are defined from the frozen backbone outputs. Directional ordering (GT $\approx$ FP < Benign for ADC; GT < FP < Benign for T2W) was preserved in all 5 seeds, replicating the PI-CAI finding.

Supplementary Tables

Table S-5fold-Summary. Five-fold cross-validation summary for P2a (case-level specificity refinement) and P2b (evidence-grounded false-positive analysis) across seven PI-CAI scenarios and the Prostate158 external cohort. Each PI-CAI row aggregates 15 observations (5 folds x 3 seeds: 42, 123, 789); the Prostate158 row aggregates 15 observations on the held-out cohort.

Scenario	P2a CaseSpec mean +/- SD (n = 15)	P2a Delta vs paper-reference	P2b MAE mean +/- SD (n = 15)	P2b lesion > benign reproduction	P2b margin (lesion - benign) mean +/- SD
PI-CAI ideal	0.4926 +/- 0.150	-0.057 vs fold-0 5-seed (0.549)	0.344 +/- 0.014	15 / 15	0.297 +/- 0.035
missing_t2w	0.1363 +/- 0.105	(no fold-0 5-seed reference)	0.426 +/- 0.017	15 / 15	0.247 +/- 0.057
missing_hbv	0.4758 +/- 0.152	–	0.405 +/- 0.015	15 / 15	0.155 +/- 0.023
missing_adc	0.4808 +/- 0.131	–	0.381 +/- 0.013	15 / 15	0.268 +/- 0.027
artifact_t2w	0.6627 +/- 0.185	–	0.347 +/- 0.015	15 / 15	0.278 +/- 0.031
artifact_hbv	0.4923 +/- 0.156	–	0.347 +/- 0.013	15 / 15	0.287 +/- 0.037
artifact_adc	0.4486 +/- 0.144	–	0.348 +/- 0.012	15 / 15	0.287 +/- 0.032
Prostate158	0.0036 +/- 0.010	(architecture- conditional collapse)	0.373 +/- 0.029	(cross-cohort; see Table 5)	(cross-cohort; see Table 5)

Footnote. P2a Delta vs paper-reference reports the difference relative to the Paper 2 fold-0 5-seed reference (0.549) on PI-CAI ideal; the remaining six PI-CAI scenarios do not have a 5-seed reference in this manuscript. P2b “lesion > benign reproduction” tallies the number of (fold, seed) observations in which the evidence-head lesion mean exceeded the benign mean. Across the seven PI-CAI scenarios x 15 observations, lesion > benign reproduces in 105 / 105. Source: 5-fold audit-dump summaries (p2a_5fold_summary.json, p2b_5fold_summary.json).

Table S-5fold-PerFold. Per-(fold, seed) P2a case-level specificity on PI-CAI ideal scenario. Comparison against the Paper 1 A2 backbone 5-fold case-level specificity baseline (0.4562). Bold per-fold mean = match-or-exceed the Paper 2 fold-0 5-seed reference (+17.2% on PI-CAI fold-0, CaseSpec 0.549; equivalently, fold mean ≥ 0.5347).

Fold	Seed 42	Seed 123	Seed 789	Fold mean	Δ vs A2 baseline (0.4562)	Relative
0	0.6296	0.4444	0.5648	0.5463	+0.0901	+19.7%
1	0.5619	0.5841	0.5531	0.5664	+0.1102	+24.1%
2	0.3272	0.5484	0.3641	0.4132	−0.0430	−9.4%
3	0.3921	0.6079	0.7489	0.5830	+0.1268	+27.8%
4	0.3810	0.5381	0.1429	0.3540	−0.1022	−22.4%

Footnote. Per-seed CaseSpec extracted from the 15 per-(fold, seed) result files (`p2a_fold*_seed*_picai.json`; computed fold means match the audit-dump summary `p2a_5fold_summary.json` to four decimal places). Seeds used in the 5-fold supplementary were 42, 123, 789 (the original Paper 2 fold-0 reference used seeds 42, 123, 456, 789, 1024). The 0.4562 baseline is the Paper 1 A2 5-fold mean case-level specificity. Direction sign across all 15 observations: 9 / 15 positive, 6 / 15 negative; range $[-0.143$ (fold 4 seed 789, relative -31.3%) to $+0.293$ (fold 3 seed 789, relative $+64.2\%$)] versus the per-fold A2 baseline.

Table S-35backbones. Backbone-agnostic verification of the P2b false-positive contrast-matching mechanism. The context-aware evidence head (identical architecture and hyperparameters) was trained on five backbones across all 5 PI-CAI folds. Observation counts: the four added backbones (bare nnU-Net, bare U-Net, bare Mamba, MIGF-Mamba A2) contribute 20 observations (4 backbones x 5 folds at seed 42); MIGF-nnUNet A2 contributes its full 15-observation reference (5 folds x 3 seeds: 42, 123, 789); total = 35. For readability the sub-tables below first group the seed-42 rows of all five backbones (a 25-observation block), then add the MIGF-nnUNet A2 seed 123 and 789 supplement (10 observations), summing to 35. Both directions reproduce in 35 / 35 observations across five distinct architectures.

Sub-table A. Lesion-evidence > benign-evidence reproduction.

Backbone	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4	Reproductions / 5
bare nnU-Net	1	1	1	1	1	5/5
bare U-Net	1	1	1	1	1	5/5
bare Mamba	1	1	1	1	1	5/5
MIGF-Mamba A2	1	1	1	1	1	5/5
MIGF-nnUNet A2	1	1	1	1	1	5/5
(seed 42 subset)						
Sub-total						25/25
MIGF-nnUNet A2						10/10
(seeds 123, 789 supplement)						
TOTAL						35/35

Sub-table B. False-positive-evidence > benign-evidence reproduction.

Backbone	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4	Reproductions / 5
bare nnU-Net	1	1	1	1	1	5/5
bare U-Net	1	1	1	1	1	5/5
bare Mamba	1	1	1	1	1	5/5
MIGF-Mamba A2	1	1	1	1	1	5/5
MIGF-nnUNet A2	1	1	1	1	1	5/5
(seed 42 subset)						
Sub-total						25/25
MIGF-nnUNet A2						10/10
(seeds 123, 789 supplement)						
TOTAL						35/35

Footnote. The four added backbones (bare nnU-Net, bare U-Net, bare Mamba, MIGF-Mamba A2) contribute 20 observations (4 backbones x 5 folds x seed 42); MIGF-nnUNet A2 contributes its full 5-fold x 3-seed P2b reference of 15 observations. Combined, 35 / 35 observations replicate both directions (lesion > benign and false-positive > benign). Cohen’s d on lesion-vs-benign margin pooled across MIGF-nnUNet A2 5-fold: 1.10 (range 0.91-1.36). FP / benign evidence ratio mean: 2.38x (range 1.91x-2.69x). The P2b context-aware evidence head, hyperparameters, and 100-epoch

Adam ($\text{lr} = 1\text{e-}4$) training protocol are identical across all five backbones. Implication: the false-positive contrast-matching phenomenon (false positives sharing raw imaging contrast with true cancer) is a data-level imaging property of prostate MRI, not a backbone-specific artifact.

Table S-Baselines. Per-seed case-level specificity at matched case-level sensitivity (target ≈ 0.94) for three post-hoc baseline strategies versus the P2a refinement head, on PI-CAI fold-0 validation ($n = 300$) and Prostate158 ($n = 158$).

Method	PI-CAI					Prostate158					Prostate158	
	s42	s123	s456	s789	s1024	Mean $\pm$ SD	s42	s123	s456	s789	s1024	Mean $\pm$ SD
Temperature scaling	0.042	0.000	0.000	0.000	0.000	0.008 $\pm$ 0.019	0.026	—	—	—	—	0.026 (n = 1)
Platt scaling	0.042	0.069	0.146	0.139	0.097	0.099 $\pm$ 0.045	0.026	—	—	—	—	0.026 (n = 1)
Threshold sweep	0.023	0.060	0.000	0.051	0.051	0.037 $\pm$ 0.025	0.089	0.036	0.018	0.089	0.054	0.057 $\pm$ 0.032
P2a refinement	0.042	0.106	0.000	0.106	0.120	0.075 $\pm$ 0.052	0.125	0.054	0.107	0.125	0.036	0.089 $\pm$ 0.042

Footnote. At matched sensitivity (target 0.94 ± 0.02). Temperature and Platt scaling are shown here for completeness; under our case-level decision rule (connected component ≥ 10 voxels), monotonic recalibration preserves the ROC and yields identical CaseSpec to threshold sweep. Temperature and Platt scaling values on Prostate158 are single-seed (seed 42) because the remaining four seeds degenerate to $T \rightarrow \infty$ or produce ROC-equivalent operating points (see experiments/baselines/DECISION.md). Per-seed deltas (P2a – threshold sweep): PI-CAI +0.019 / +0.046 / 0.000 / +0.056 / +0.069 (mean +0.038 $\pm$ 0.028); Prostate158 +0.036 / +0.018 / 0.000 / –0.018 / +0.089 (mean +0.032 $\pm$ 0.039). Seed 456 on PI-CAI is the saturated all-positive threshold seed for which no method can reduce false positives without violating matched sensitivity.

Supplementary Table S3. Case-level specificity on Prostate158 at four discrete sensitivity targets, for the bare backbone (A2) and the refined model (P2a, multi-scale v2), 5 seeds. “Reached” means the per-seed operating curve achieved case-level sensitivity within ± 0.02 of the target. Values in the “mean CaseSpec (reached)” columns are 5-seed mean $\pm$ sample standard deviation (ddof = 1) restricted to seeds that reached the target; “n/5 reached” reports the number of seeds. Near-saturation targets (0.90, 0.95) produce sparse reached counts on both models.

Target CaseSens	Bare CaseSpec (reached)	Bare n/5 reached	Refined CaseSpec (reached)	Refined n/5 reached
0.80	0.25	1/5	0.28 ± 0.01	2/5
0.85	0.18	1/5	0.24 ± 0.04	2/5
0.90	0.13 ± 0.03	2/5	0.11 ± 0.08	2/5
0.95	0.06 ± 0.03	3/5	0.07 ± 0.03	4/5

Source: `experiments/prostate158_sensspec_curve/sensspec_aggregate.json`, 2026-04-20.

Table S-Stat. Seed-level paired Wilcoxon signed-rank analysis of matched-sensitivity case-level specificity gain (P2a – bare A2), retained as a sensitivity check to the main-text case-level inference (§3.7).

Cohort	Metric	5-seed per-seed values	Mean $\pm$ SD	Wilcoxon signed-rank one-sided p (H1: P2a > bare)
PI-CAI	Δ CaseSpec at	+0.019, +0.046, 0.000,	+0.038	p = 0.06
fold-0 val	matched sens	+0.056, +0.069	$\pm$ 0.028	
Prostate158	Δ CaseSpec at	+0.036, +0.018, 0.000,	−0.011	p = 0.62
	matched sens	−0.018, −0.089	$\pm$ 0.048	

Footnote. Seed-level Wilcoxon retained as sensitivity analysis; main-text inference uses case-level McNemar + bootstrap per §3.7. Per-seed Δ CaseSpec on PI-CAI from `experiments/p2a_5fold_cv/bootstrap_re` per-seed Δ CaseSpec on Prostate158 from `experiments/prostate158_matched_sens/matched_sens_aggregate`. Wilcoxon one-sided p was computed with `scipy.stats.wilcoxon(alternative="greater", zero_method="zsplit")` (handles the tied zero at PI-CAI seed 456). The PI-CAI p = 0.06 approaches but does not cross $\alpha = 0.05$ at n = 5 seeds, consistent with the n = 5 Wilcoxon structural minimum p $\approx$ 0.03 and illustrating why the main-text inference is case-level rather than seed-level.

Table S-Bootstrap. Per-seed case-level paired bootstrap confidence intervals for the matched-sensitivity case-level specificity gain (P2a – bare A2) on Prostate158 negative cases ($n = 56$ per seed). Pooled row aggregates all 5 seeds $\times$ 56 negative cases = 280 case-observations.

Seed	Cohort	n negative cases	Δ CaseSpec (per-case mean)	95% CI lower	95% CI upper	p (two-sided, bootstrap)
42	Prostate158	56	+0.036	0.000	+0.089	0.268
123	Prostate158	56	+0.018	0.000	+0.054	0.721
456	Prostate158	56	0.000	0.000	0.000	1.000
789	Prostate158	56	−0.018	−0.071	+0.036	0.782
1024	Prostate158	56	−0.089	−0.196	0.000	0.112
Pooled (5 seeds $\times$ 56)	Prostate158	280	−0.011	−0.039	+0.018	0.517

Footnote. Per-case paired bootstrap with 10,000 resamples (rng seed 20260421). Seed-level estimates pooled across all 5 seeds $\times$ 56 negative cases = 280 case-observations; the pooled resampler draws with replacement from the pooled (seed, case) observation index. The PI-CAI per-case detection sidecars required to run the analogous analysis on PI-CAI fold-0 validation were not available in the local experiments tree; the PI-CAI bootstrap reported in the main text (§3.7) is therefore the non-paired per-seed cohort resampling from `experiments/p2a_5fold_cv/bootstrap_results.json`. Seed 456 is the saturated all-positive seed for which $\Delta = 0$ by construction.

Table S-Stratification. Pooled-across-5-seeds false-positive counts per similarity tier and per-tier suppression rates for PI-CAI fold-0 validation and Prostate158. Tiers defined by tertiles of cosine similarity between the FP-region contrast-ratio vector and the case-specific ground-truth cancer contrast-ratio vector. Negative-case FPs (FPs in patients with no ground-truth cancer) have no case-specific cancer vector and are reported separately.

Cohort	Tier	n FPs (5-seed pooled)	% of total	Suppression rate
PI-CAI	High similarity	1,401	9.8%	27.7%
PI-CAI	Mid similarity	1,401	9.8%	25.9%
PI-CAI	Low similarity	1,401	9.8%	30.5%
PI-CAI	Negative-case FPs	10,071	70.5%	28.1%
Prostate158	High similarity	1,494	22.0%	28.6%
Prostate158	Mid similarity	1,494	22.0%	28.4%
Prostate158	Low similarity	1,494	22.0%	28.0%
Prostate158	Negative-case FPs	2,313	34.0%	29.1%

Footnote. Tiers defined by tertiles of cosine similarity between FP-region contrast-ratio vector and case-specific GT cancer contrast-ratio vector. Uniform suppression across tiers (within ± 2 percentage points of the cohort overall rate of 28.1% on PI-CAI and 28.6% on Prostate158); a χ^2 test of independence between tier and suppressed/residual status was non-significant for both cohorts. Negative-case FPs are enumerated separately because no case-specific cancer vector exists for those cases. Tertile boundaries: PI-CAI p33 = 0.151, p67 = 0.764; Prostate158 p33 = 0.069, p67 = 0.816.